

PROBLEM STATEMENT

Agricultural Equipment Rental Marketplace for Farmers

1. Title

Agricultural Equipment Rental Marketplace for Farmers

2. Domain

Agriculture / AgriTech / Full-Stack Web Application

3. Who is the User?

Farmer

- Registers and logs in.
- Browses and searches equipment.
- Books equipment for a required period.
- Manages rental bookings.

Equipment Owner

- Registers and logs in.
- Lists equipment for rental.
- Updates equipment and availability.
- Accepts or rejects booking requests.

Admin

- Manages users and listings.
- Monitors bookings and activities.
- Handles invalid listings.

4. What Problem Are We Solving?

Many small and medium-scale farmers cannot afford expensive agricultural equipment such as tractors, harvesters, sprayers, and seed drills. These machines may only be needed during particular farming seasons, making ownership costly and inefficient. At the same time, equipment owners may have machinery that remains unused for long periods. For example, a farmer who needs a tractor for only two days should not have to purchase an expensive tractor. The proposed platform connects farmers with equipment owners so farmers can easily find and rent the required machinery.

5. Proposed Solution

- User registration and secure login.
- Separate access for farmers, equipment owners, and administrators.
- Equipment owners can add, edit, delete, and manage listings.
- Farmers can browse, search, and filter available equipment.
- Farmers can select rental dates and submit booking requests.
- Equipment owners can approve or reject booking requests.
- Farmers can view booking status and history.
- Users can manage profiles.
- Database stores users, equipment, bookings, and related information.
- Cloud deployment provides online access.

6. Core Entities / Database Tables

- Users – account and login information.
- Farmers – farmer-specific information.
- Equipment_Owners – equipment owner information.
- Equipment – equipment details, rental price, category, and availability.
- Bookings – rental booking details and status.
- Payments – payment information, if implemented.
- Reviews – ratings and reviews.

7. User Roles & Permissions

Role	Permissions
Farmer	Register, login, browse/search, book equipment, view/cancel bookings, manage profile.
Equipment Owner	Register, login, add/update/remove equipment, manage availability, approve/reject bookings.
Admin	Manage users, monitor listings and bookings, manage platform activities.

8. Success Criteria

- Users can register and log in successfully.
- Farmers can find available equipment easily.
- A farmer can create a rental booking within a few minutes.
- Equipment owners can manage listings and booking requests.
- Users can view current and previous bookings.
- Data is correctly stored and retrieved from the database.
- The application works correctly after cloud deployment.

9. Out of Scope

- Equipment transportation or delivery.
- Maintenance and repair services.
- Insurance management.
- Real-time GPS tracking.
- IoT-based equipment monitoring.
- Automatic rental price prediction.
- International rental operations.
- Advanced payment services unless specifically required.
- Mobile application development; the initial version is a web application.

10. Chosen Track

Java – Spring Boot

Technology Stack

- Frontend: HTML, CSS, JavaScript / React
- Backend: Java + Spring Boot

- Database: MySQL
- Database Access: Spring Data JPA / Hibernate
- API: REST API
- Version Control: Git + GitHub
- Development Environment: VS Code
- Deployment: AWS / Google Cloud / Render